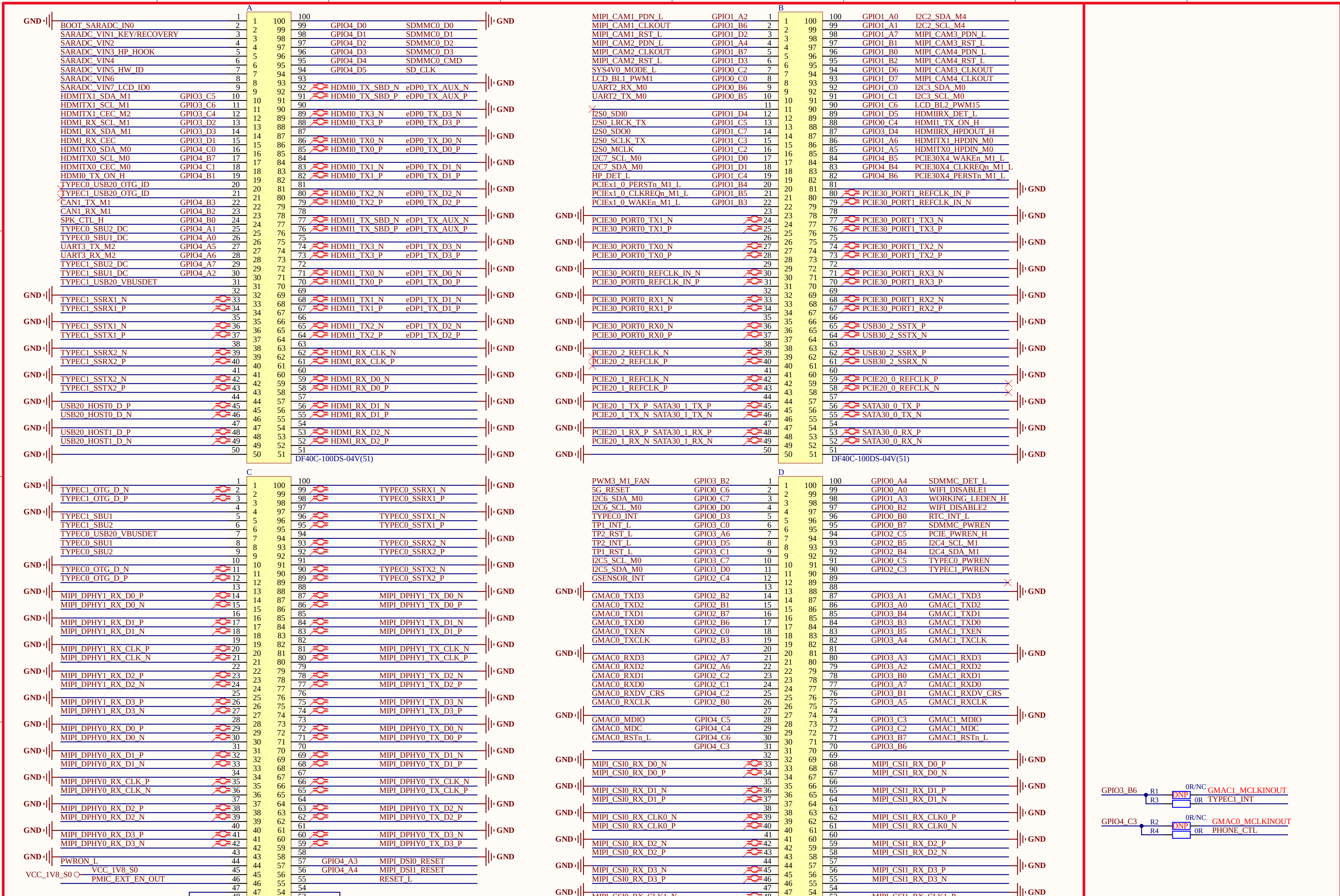
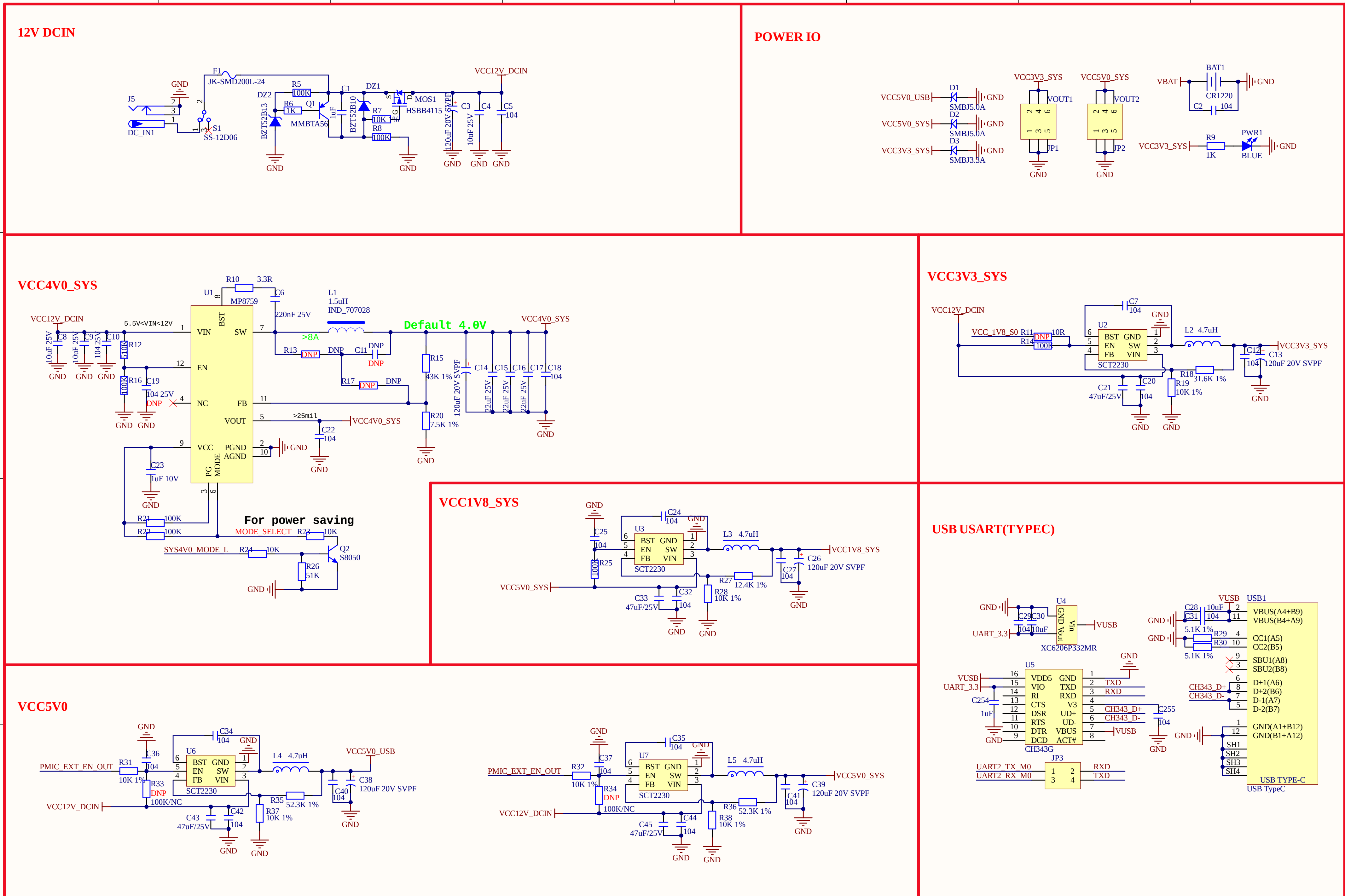
Revision History

Version	Date	By	Change Dscription
V1.0	2024-04-23	zuozhongkai	1:Initial release version
V1.1	2025-04-11	zuozhongkai	1:R1，R2改为NC，默认不贴片

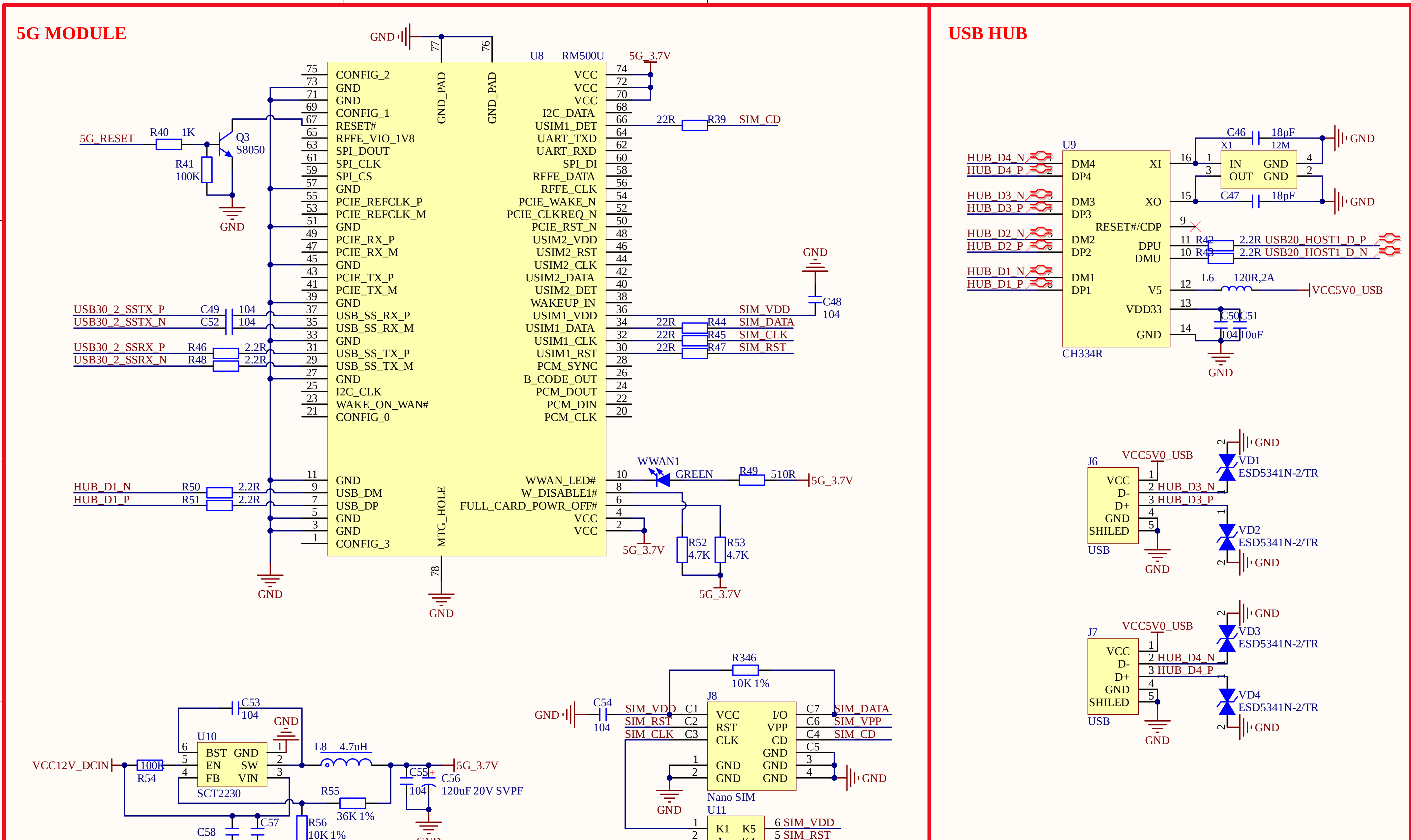


Title: 02_ATK_RK3588_Core_Board.SchDoc	
Project: ATK-DLRK3588B V1.1.PrjPcb	
Size: A3	Author: ALIENTEK
Date: 2025/9/5	Version: V1.1 Sheet: 2 of 17

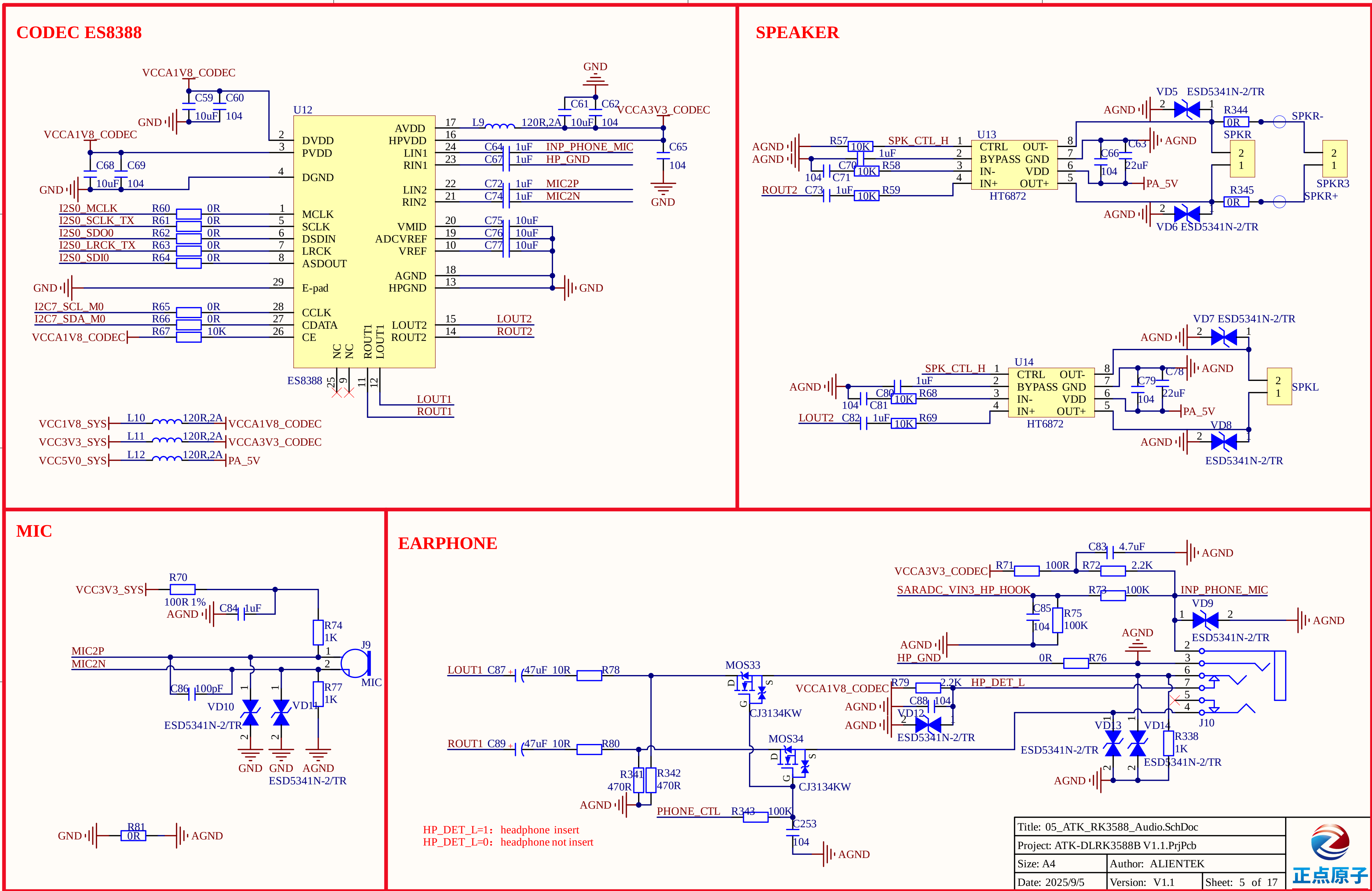
正点原子



Title: 03_ATK_RK3588_Power_USB.SchDoc	
Project: ATK-DLRK3588B V1.1.PrjPcb	
Size: A3	Author: ALIENTEK
Date: 2025/9/5	Version: V1.1
Sheet: 3 of 17	



Title: 04_ATK_RK3588_5G_HUB.SchDoc		
Project: ATK-DLRK3588B V1.1.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/9/5	Version: V1.1	Sheet: 4 of 17



Title: 05_ATK_RK3588_Audio.SchDoc	
Project: ATK-DLRK3588B V1.1.PrjPcb	
Size: A4	Author: ALIENTEK
Date: 2025/9/5	Version: V1.1
Sheet: 5 of 17	

正点原子

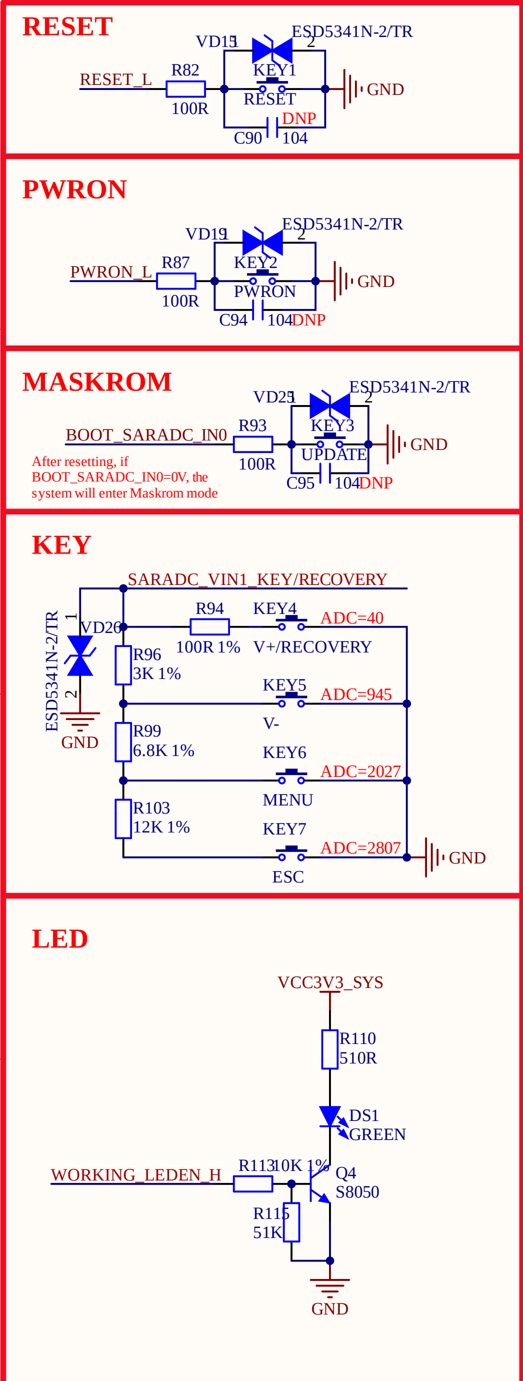
RESET

PWRON

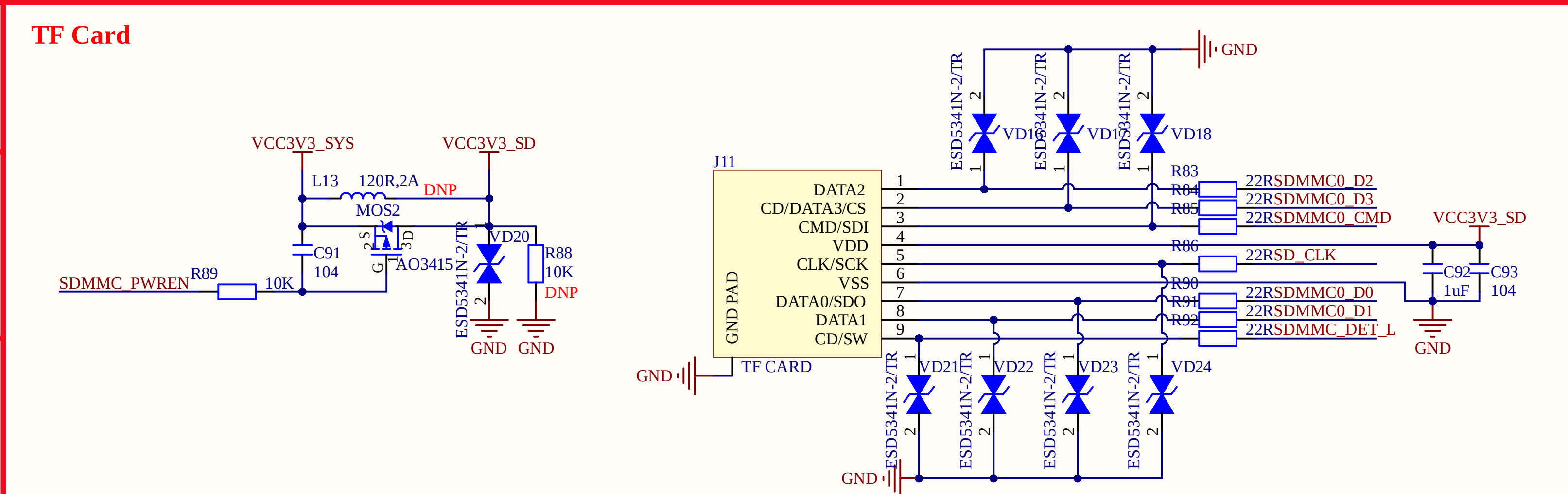
MASKROM

KEY

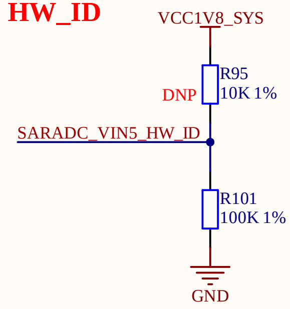
LED



TF Card

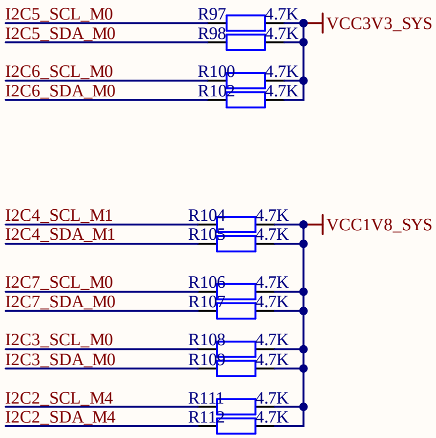


HW_ID

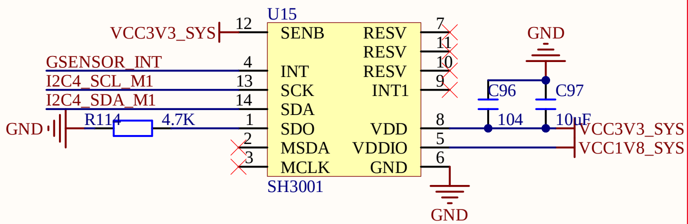


HW_ID	Rup	Rdown	ADC	
LEVEL1	DNP	100K	0	0V
LEVEL2	100K	20K	682	0.3V
LEVEL3	100K	51K	1365	0.6V
LEVEL4	100K	100K	2047	0.9V
LEVEL5	100K	200K	2730	1.2V
LEVEL6	100K	499K	3412	1.5V
LEVEL7	100K	DNP	4095	1.8V

IIC



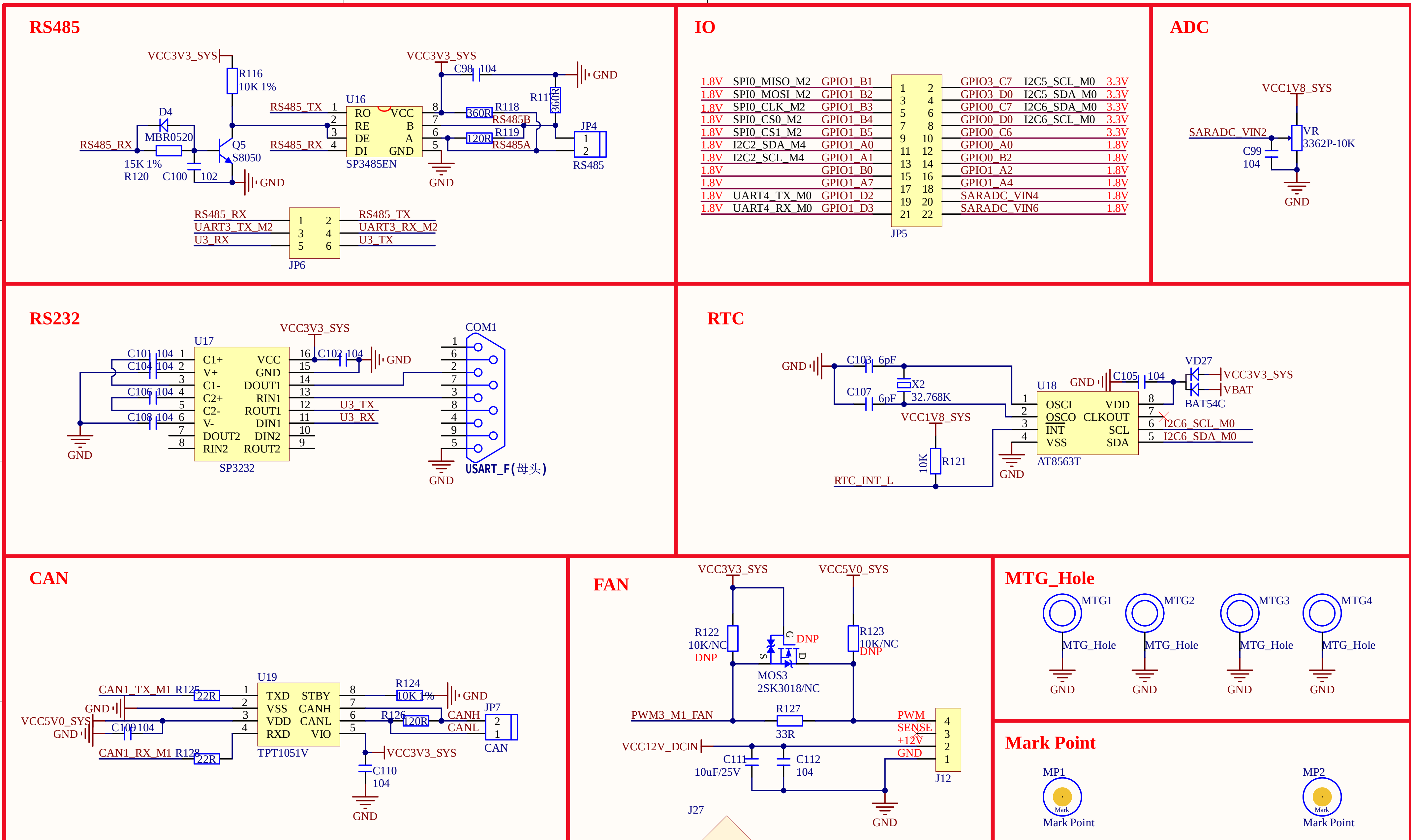
6 AXIS SENSOR



SH3001 IIC Address:0X36

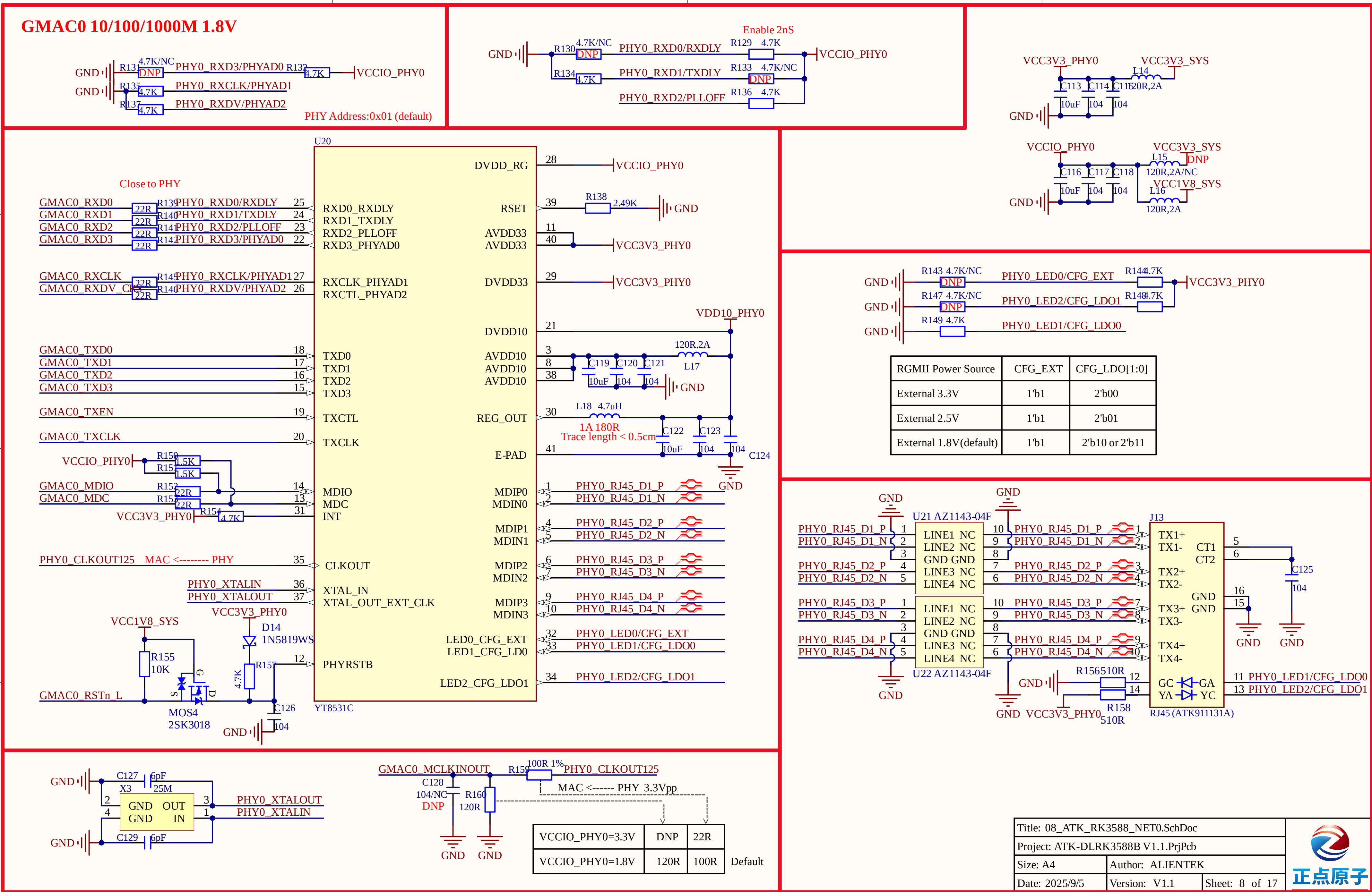
Title: 06_ATK_RK3588_DEVICE1.SchDoc	
Project: ATK-DLRK3588B V1.1.PrjPcb	
Size: A4	Author: ALIENTEK
Date: 2025/9/5	Version: V1.1
Sheet: 6 of 17	

正点原子



Title: 07_ATK_RK3588_DEVICE2.SchDoc		
Project: ATK-DLRK3588B V1.1.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/9/5	Version: V1.1	Sheet: 7 of 17

GMAC0 10/100/1000M 1.8V

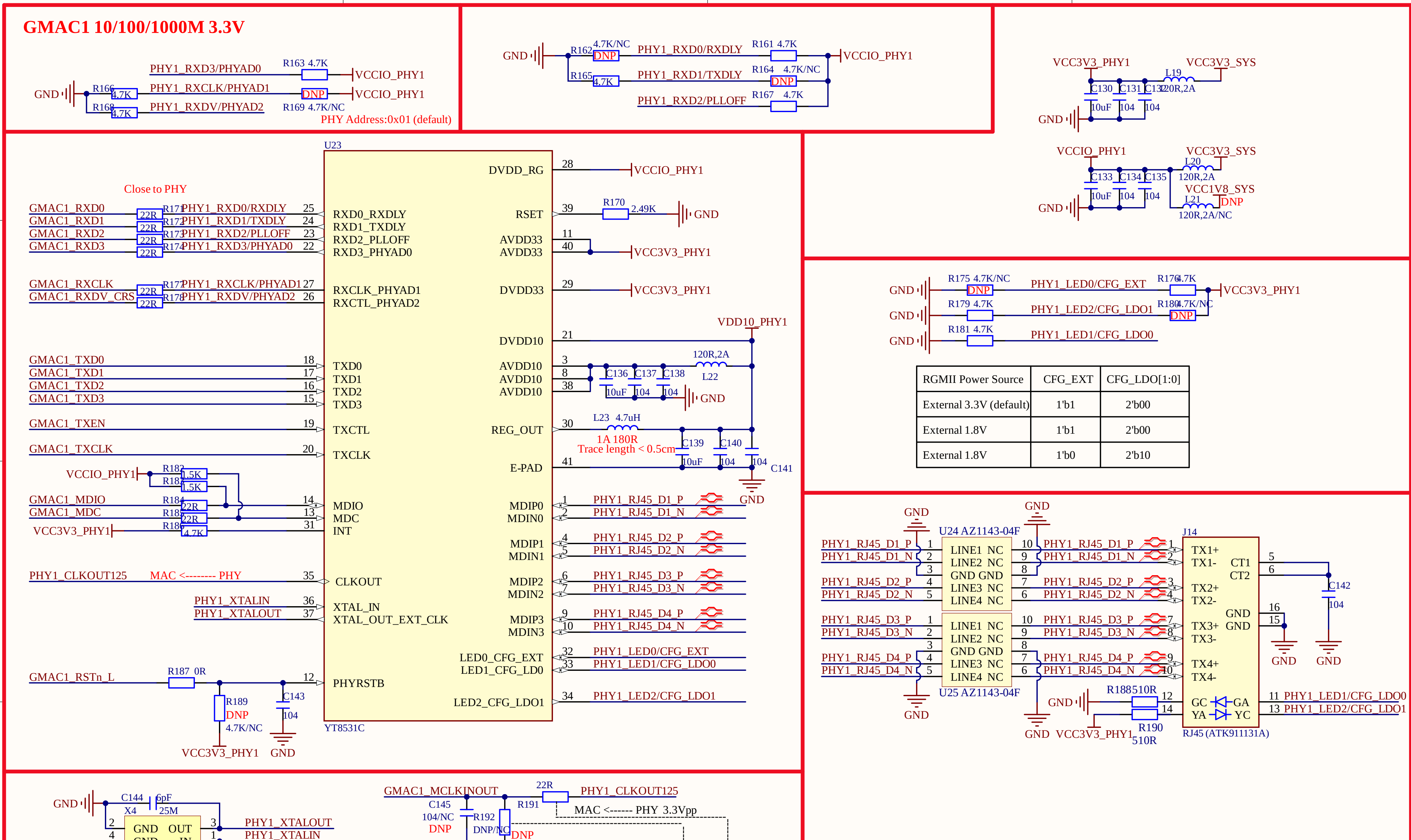


RGMII Power Source	CFG_EXT	CFG_LDO[1:0]
External 3.3V	1'b1	2'b00
External 2.5V	1'b1	2'b01
External 1.8V(default)	1'b1	2'b10 or 2'b11

VCCIO_PHY0=3.3V	DNP	22R	
VCCIO_PHY0=1.8V	120R	100R	Default

Title: 08_ATK_RK3588_NET0.SchDoc
Project: ATK-DLRK3588B V1.1.PrjPcb
Size: A4 | Author: ALIENTEK
Date: 2025/9/5 | Version: V1.1 | Sheet: 8 of 17

正点原子

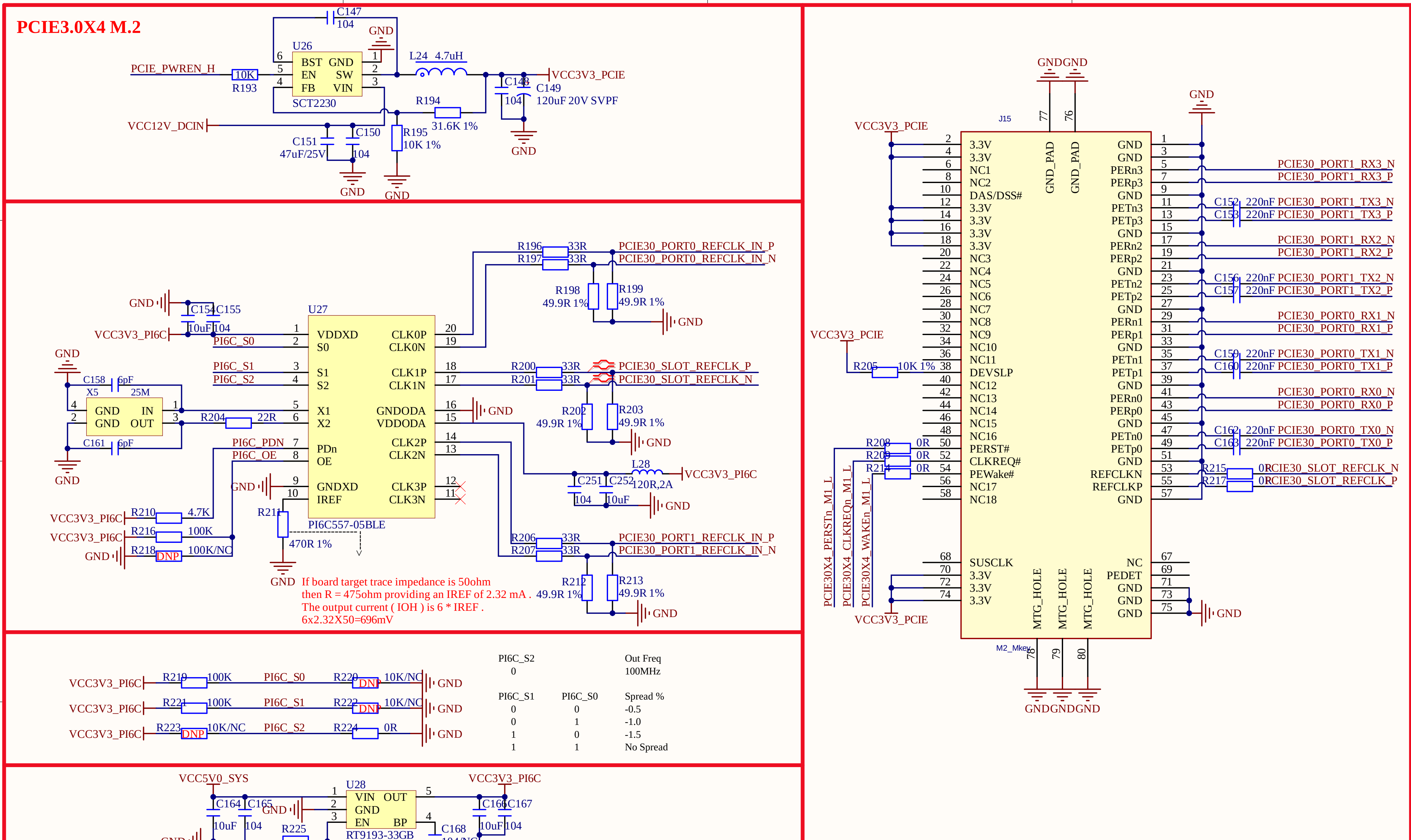


RGMII Power Source	CFG_EXT	CFG_LDO[1:0]
External 3.3V (default)	1'b1	2'b00
External 1.8V	1'b1	2'b00
External 1.8V	1'b0	2'b10

VCCIO_PHY0=3.3V	DNP	22R	Default
VCCIO_PHY0=1.8V	120R	100R	

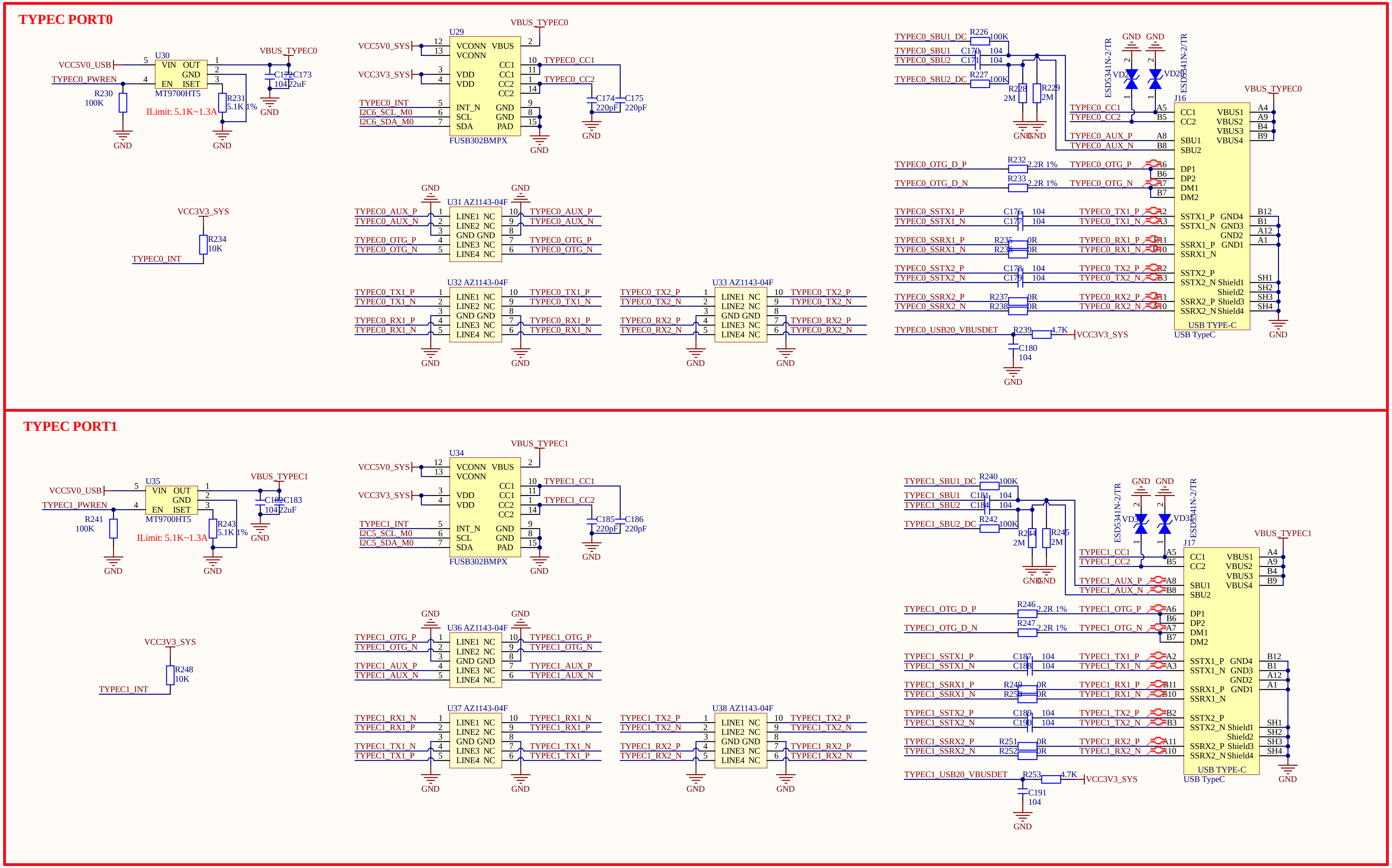
Title: 09_ATK_RK3588_NET1.SchDoc		
Project: ATK-DLRK3588B V1.1.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/9/5	Version: V1.1	Sheet: 9 of 17

正点原子



Title: 10_ATK_RK3588_PCIE3.0X4_M.2.SchDoc	
Project: ATK-DLRK3588B V1.1.PrjPcb	
Size: A4	Author: ALIENTEK
Date: 2025/9/5	Version: V1.1 — Sheet: 10 of 17

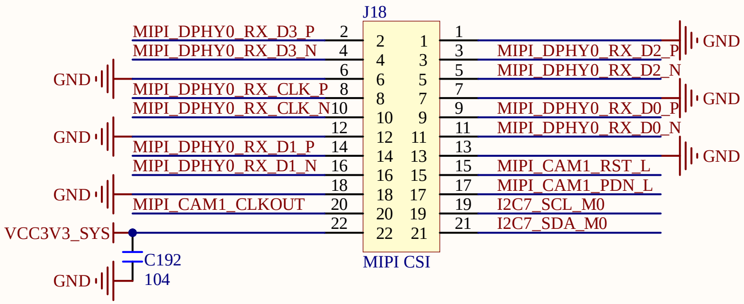
正点原子



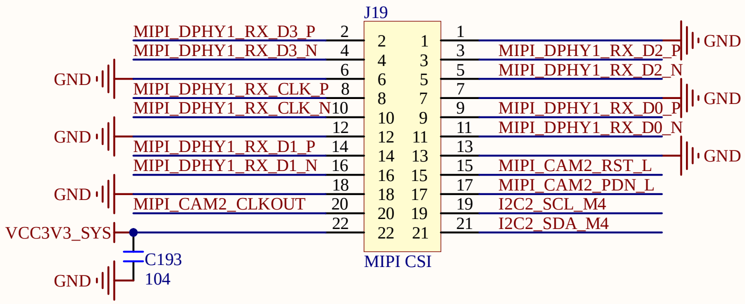
Title: 11_ATK_RK3588_TYPEC.SchDoc		
Project: ATK-DLRK3588B V1.1.PrjPcb		
Size: A3	Author: ALIENTEK	
Date: 2025/9/5	Version: V1.1	Sheet: 11 of 17

正点原子

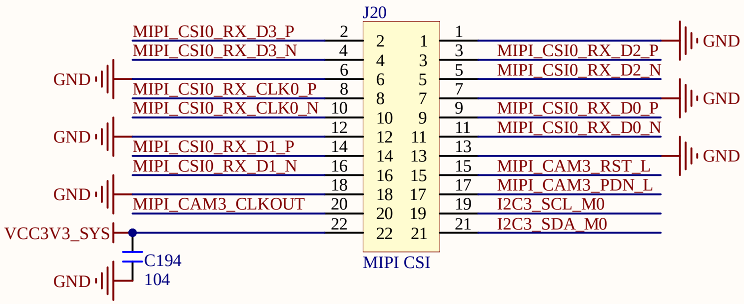
MIPI CSI1



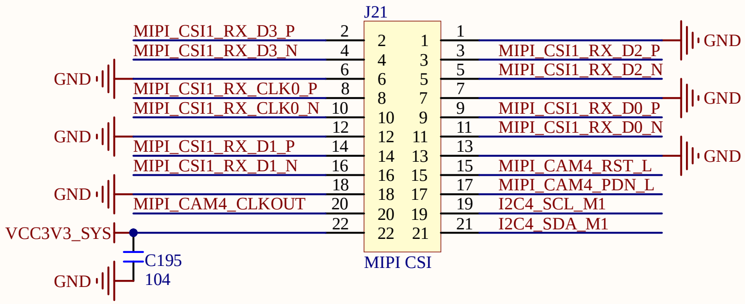
MIPI CSI2



MIPI CSI3

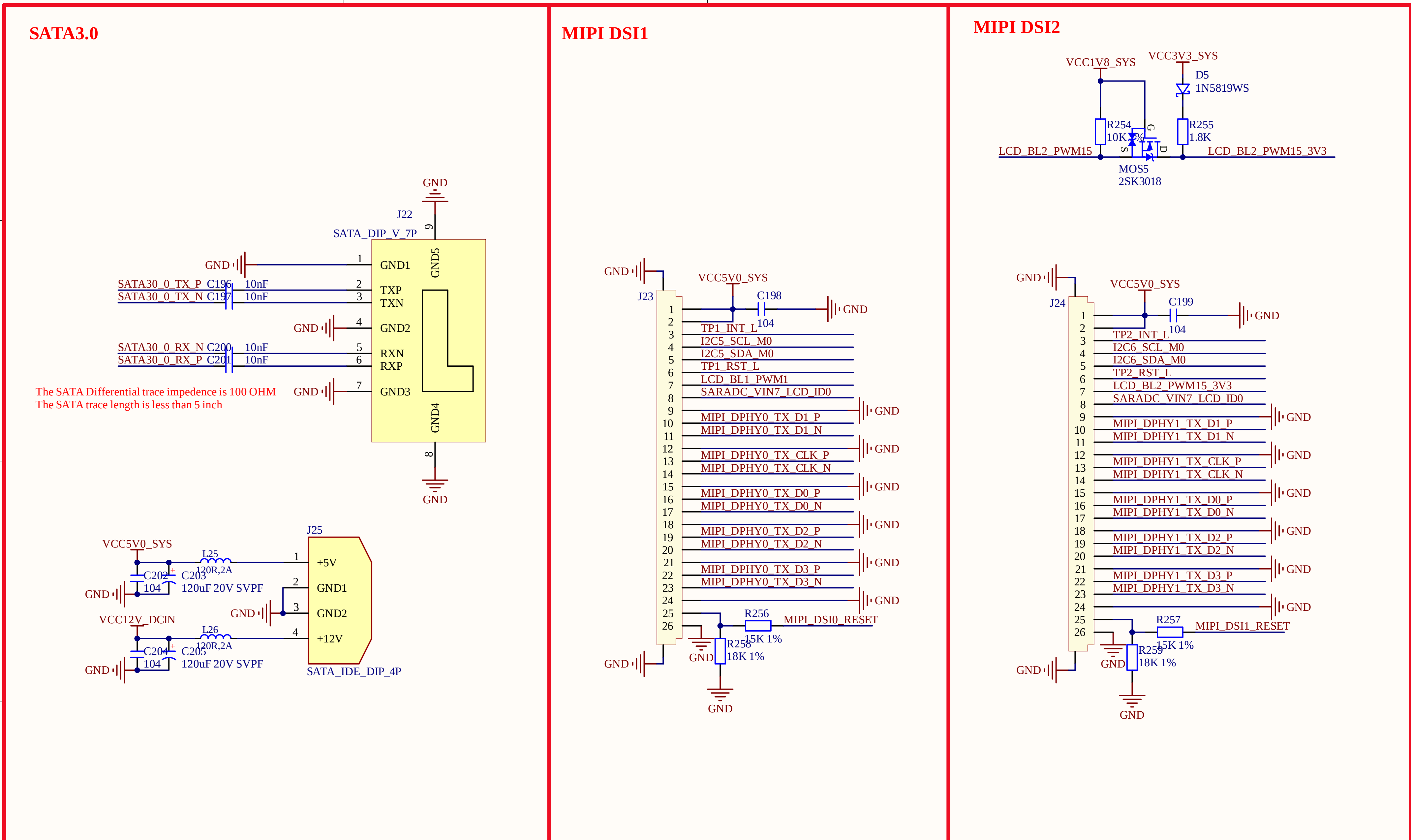


MIPI CSI4



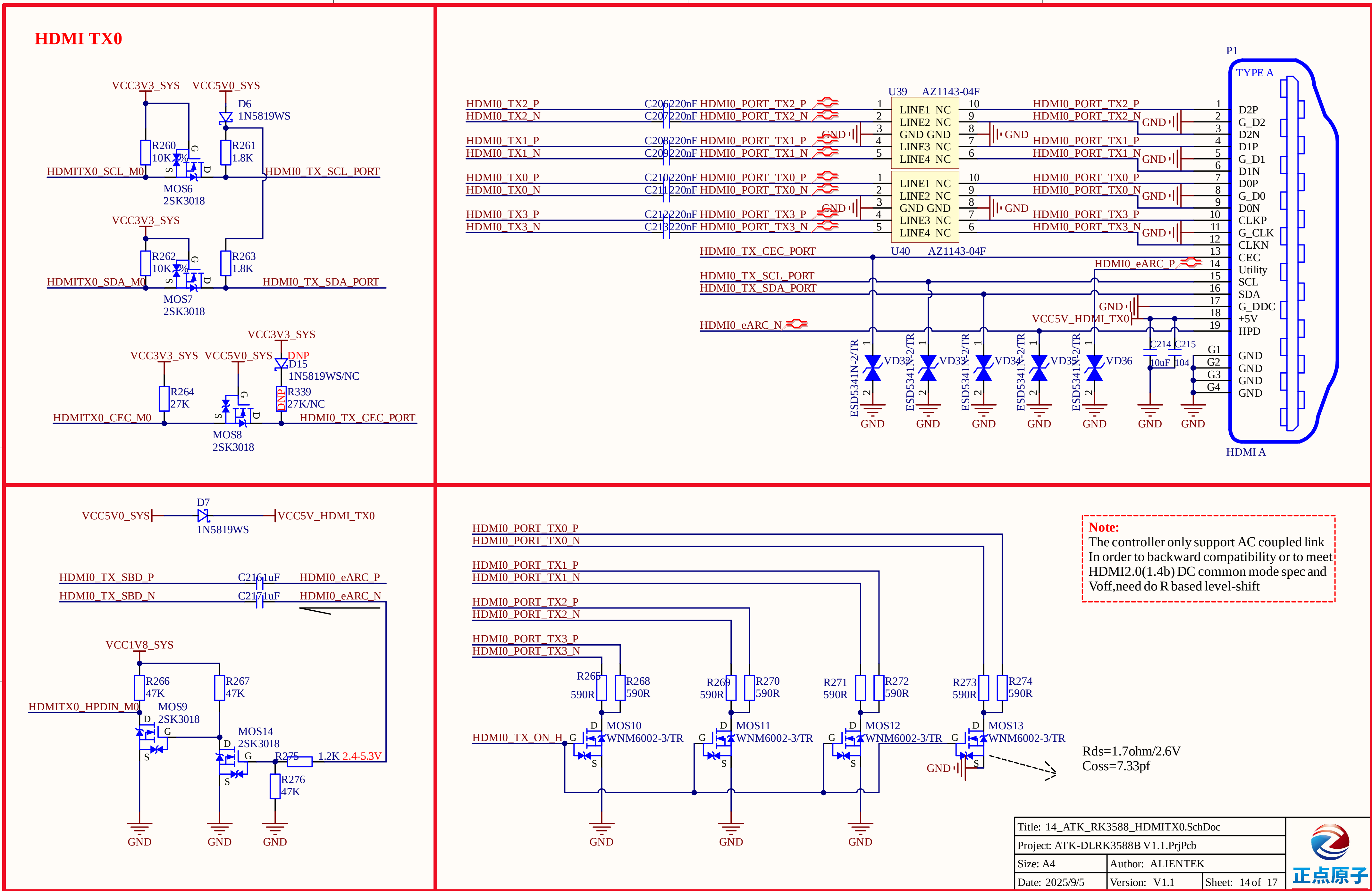
Title: 12_ATK_RK3588_CSI.SchDoc		
Project: ATK-DLRK3588B V1.1.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/9/5	Version: V1.1	Sheet: 12 of 17

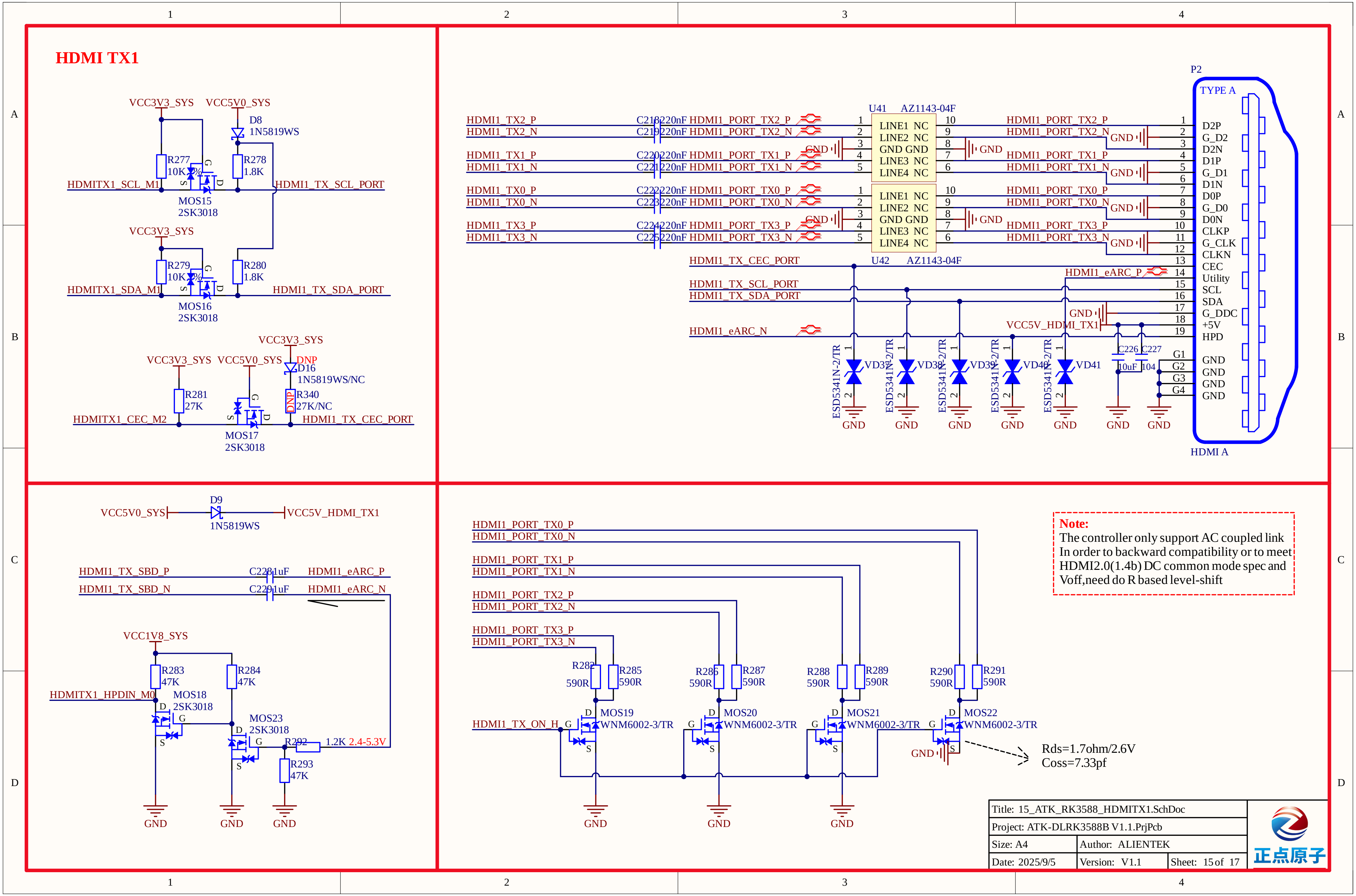
正点原子

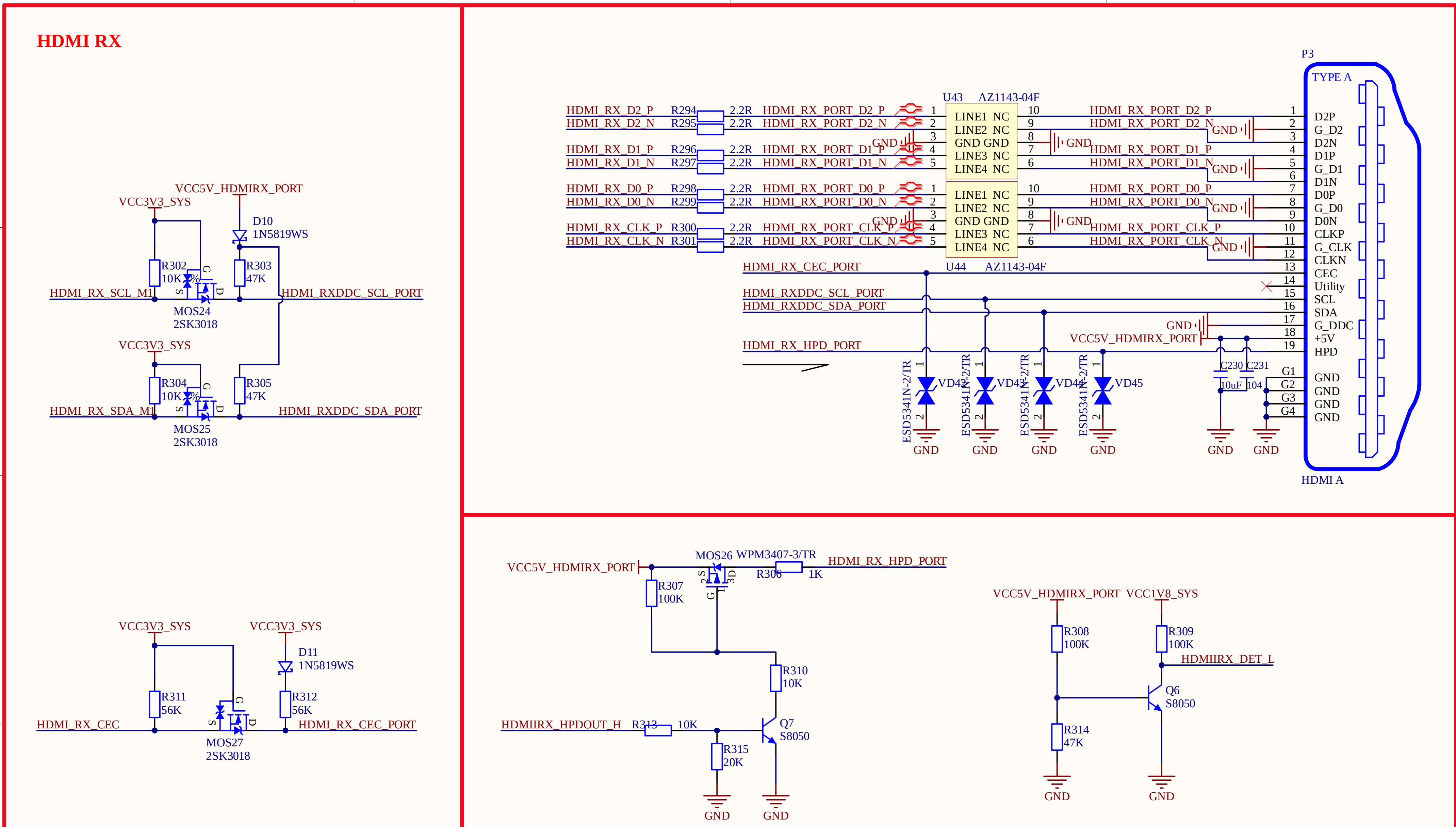


Title: 13_ATK_RK3588_SATA_DSI.SchDoc		
Project: ATK-DLRK3588B V1.1.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/9/5	Version: V1.1	Sheet: 13 of 17

正点原子


HDMI TX0
VCC3V3_SYS
VCC5V0_SYS
D6
1N5819WS
R260
10K
R261
1.8K
HDMITX0_SCL_M0
HDMI0_TX_SCL_PORT
MOS6
2SK3018
R262
10K
R263
1.8K
HDMITX0_SDA_M0
HDMI0_TX_SDA_PORT
MOS7
2SK3018
DNP
D15
1N5819WS/NC
R264
27K
R339
27K/NC
HDMITX0_CEC_M0
HDMI0_TX_CEC_PORT
MOS8
2SK3018
HDMI0_TX2_P
HDMI0_TX2_N
HDMI0_TX1_P
HDMI0_TX1_N
HDMI0_TX0_P
HDMI0_TX0_N
HDMI0_TX3_P
HDMI0_TX3_N
C206 220nF
C207 220nF
C208 220nF
C209 220nF
C210 220nF
C211 220nF
C212 220nF
C213 220nF
HDMI0_PORT_TX2_P
HDMI0_PORT_TX2_N
HDMI0_PORT_TX1_P
HDMI0_PORT_TX1_N
HDMI0_PORT_TX0_P
HDMI0_PORT_TX0_N
HDMI0_PORT_TX3_P
HDMI0_PORT_TX3_N
U39 AZ1143-04F
U40 AZ1143-04F
LINE1 NC
LINE2 NC
GND GND
LINE3 NC
LINE4 NC
GND
HDMI0_TX_CEC_PORT
HDMI0_TX_SCL_PORT
HDMI0_TX_SDA_PORT
HDMI0_eARC_P
HDMI0_eARC_N
VCC5V_HDMI_TX0
ESD5341N-2/TR
VD32
VD33
VD34
VD35
VD36
C214 10uF
C215 104
P1
TYPE A
D2P
G_D2
D2N
D1P
G_D1
D1N
D0P
G_D0
D0N
CLKP
G_CLK
CLKN
CEC
Utility
SCL
SDA
G_DDC
+5V
HPD
G1 G2 G3 G4
HDMI A
VCC5V0_SYS
D7
1N5819WS
VCC5V_HDMI_TX0
HDMI0_TX_SBD_P
HDMI0_TX_SBD_N
C216 1uF
C217 1uF
HDMI0_eARC_P
HDMI0_eARC_N
VCC1V8_SYS
R266
47K
R267
47K
HDMITX0_HPDIN_M0
MOS9
2SK3018
MOS14
2SK3018
R275
1.2K 2.4-5.3V
R276
47K
R265 590R
R268 590R
R269 590R
R270 590R
R271 590R
R272 590R
R273 590R
R274 590R
HDMI0_TX_ON_H
MOS10 WNM6002-3/TR
MOS11 WNM6002-3/TR
MOS12 WNM6002-3/TR
MOS13 WNM6002-3/TR
Note:
The controller only support AC coupled link
In order to backward compatibility or to meet
HDMI2.0(1.4b) DC common mode spec and
Voff,need do R based level-shift
Rds=1.7ohm/2.6V
Coss=7.33pf
Title: 14_ATK_RK3588_HDMITX0.SchDoc
Project: ATK-DLRK3588B V1.1.PrjPcb
Size: A4
Author: ALIENTEK
Date: 2025/9/5
Version: V1.1
Sheet: 14 of 17
正点原子

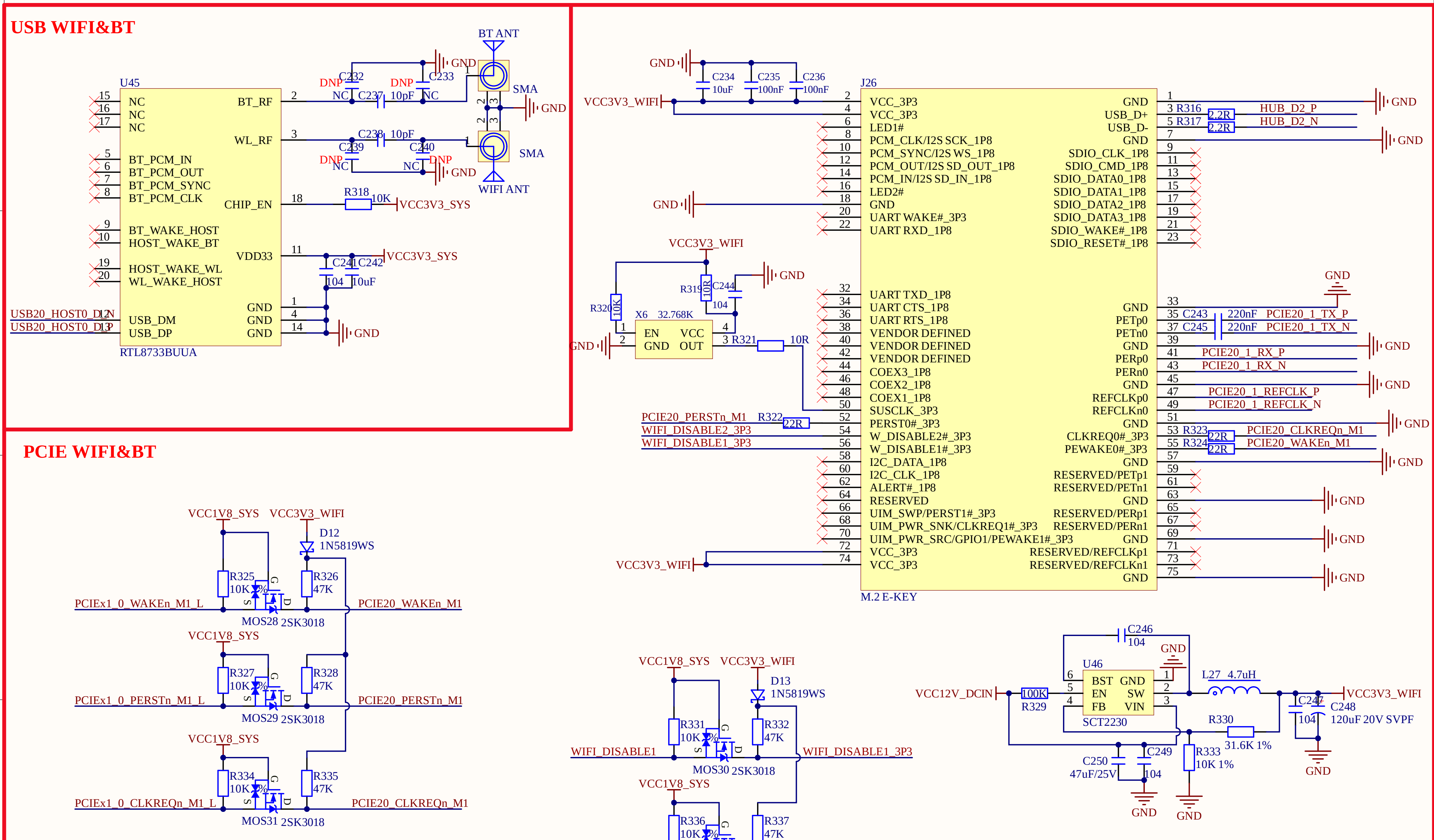

HDMI TX1
VCC3V3_SYS
VCC5V0_SYS
D8
1N5819WS
R277
10K
R278
1.8K
HDMITX1_SCL_M1
HDMI1_TX_SCL_PORT
MOS15
2SK3018
R279
10K
R280
1.8K
HDMITX1_SDA_M1
HDMI1_TX_SDA_PORT
MOS16
2SK3018
R281
27K
DNP
D16
1N5819WS/NC
R340
27K/NC
HDMITX1_CEC_M2
HDMI1_TX_CEC_PORT
MOS17
2SK3018
HDMI1_TX2_P
HDMI1_TX2_N
HDMI1_TX1_P
HDMI1_TX1_N
HDMI1_TX0_P
HDMI1_TX0_N
HDMI1_TX3_P
HDMI1_TX3_N
C218 220nF HDMI1_PORT_TX2_P
C219 220nF HDMI1_PORT_TX2_N
C220 220nF HDMI1_PORT_TX1_P
C221 220nF HDMI1_PORT_TX1_N
C222 220nF HDMI1_PORT_TX0_P
C223 220nF HDMI1_PORT_TX0_N
C224 220nF HDMI1_PORT_TX3_P
C225 220nF HDMI1_PORT_TX3_N
U41 AZ1143-04F
U42 AZ1143-04F
LINE1 NC
LINE2 NC
GND GND
LINE3 NC
LINE4 NC
GND
HDMI1_TX_CEC_PORT
HDMI1_TX_SCL_PORT
HDMI1_TX_SDA_PORT
HDMI1_eARC_N
HDMI1_eARC_P
VCC5V_HDMI_TX1
ESD5341N-2/TR
VD37
VD38
VD39
VD40
VD41
C226 10uF
C227 104
P2
TYPE A
D2P
G_D2
D2N
D1P
G_D1
D1N
D0P
G_D0
D0N
CLKP
G_CLK
CLKN
CEC
Utility
SCL
SDA
G_DDC
+5V
HPD
G1
G2
G3
G4
HDMI A
D9
1N5819WS
VCC5V0_SYS
VCC5V_HDMI_TX1
HDMI1_TX_SBD_P
C228 1uF
HDMI1_eARC_P
HDMI1_TX_SBD_N
C229 1uF
HDMI1_eARC_N
VCC1V8_SYS
R283
47K
R284
47K
HDMITX1_HPDIN_M0
MOS18
2SK3018
MOS23
2SK3018
R292
1.2K
2.4-5.3V
R293
47K
HDMI1_PORT_TX0_P
HDMI1_PORT_TX0_N
HDMI1_PORT_TX1_P
HDMI1_PORT_TX1_N
HDMI1_PORT_TX2_P
HDMI1_PORT_TX2_N
HDMI1_PORT_TX3_P
HDMI1_PORT_TX3_N
R282 590R
R285 590R
R286 590R
R287 590R
R288 590R
R289 590R
R290 590R
R291 590R
HDMI1_TX_ON_H
MOS19 WNM6002-3/TR
MOS20 WNM6002-3/TR
MOS21 WNM6002-3/TR
MOS22 WNM6002-3/TR
Rds=1.7ohm/2.6V
Coss=7.33pf
Note:
The controller only support AC coupled link
In order to backward compatibility or to meet
HDMI2.0(1.4b) DC common mode spec and
Voff,need do R based level-shift
Title: 15_ATK_RK3588_HDMITX1.SchDoc
Project: ATK-DLRK3588B V1.1.PrjPcb
Size: A4
Author: ALIENTEK
Date: 2025/9/5
Version: V1.1
Sheet: 15 of 17
正点原子



Note:
HPD:
Sink Side: Require output, Min 2.4V; Max 5.3V
Source Side: Require input and Detection.
Min 2.0V, Max:5.3V

Title: 16_ATK_RK3588_HDMIRX.SchDoc	
Project: ATK-DLRK3588B V1.1.PrjPcb	
Size: A4	Author: ALIENTEK
Date: 2025/9/5	Version: V1.1 — Sheet: 16 of 17

正点原子



Title: 17_ATK_RK3588_WIFI.SchDoc		
Project: ATK-DLRK3588B V1.1.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/9/5	Version: V1.1	Sheet: 17 of 17

正点原子